2022-2023 SPRING SEMESTER FLUID MECHANICS MIDTERM1, Date:25/04/2023

Name: **Surname:** **Student ID:**– – – – – – – – –
c b a

If a, b and c are zero in your student ID number (xxxxcba), please take it 2.

a; the last digit of your student ID number

b; the penultimate digit of your student ID number

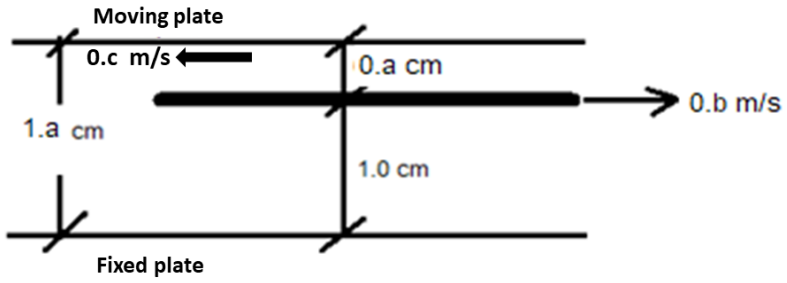
c; the second before the last digit of your student ID number

Question 1 (25p) (a)

The space between two parallel plates 1.a cm apart is filled with an oil of viscosity μ=0.0b0 kg/m.s. The lower plate is stationary, and the upper plate moves at a constant velocity of 0.c0 m/s. A thin 30x60 cm^2 rectangular plate which have same area with above plate is pulled through the oil 0.a0 cm from one plate and 1.0 cm from the other.

a) What is the magnitude of force to pull the plate at 0.b0 m/s?

b) Draw the velocity profile between the plates.



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Question 1 (b)

What is the pressure difference in the following fig?

Specific gravity of oil d=0.8a

Specific gravity of carbon tetrachloride d=1.5b

Specific gravity of mercury d=13.c

